

# YERMAN J. MEREL G.

**Mechanical Engineer | Mechanical Design Engineer | Project Engineer**

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## PROFESSIONAL SUMMARY

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Mechanical Engineer with experience in large-scale infrastructure projects, specializing in mechanical systems analysis, technical documentation, and engineering support for complex equipment.

Hands-on experience in mechanical design using Autodesk Inventor (3D modeling, assemblies, and manufacturing drawings) and finite element analysis (FEA) using Ansys for structural validation.

## CORE COMPETENCIES

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**Mechanical Design & Analysis:** 3D Modeling & Assemblies (Autodesk Inventor) • Manufacturing Drawings • Engineering Calculations • Structural Analysis • Finite Element Analysis (FEA – Ansys)

**Engineering & Technical:** Mechanical Systems Analysis • Product Support & Validation • Technical Documentation • Piping Systems (P&IDs / Isometrics – ASME B31) • Equipment Performance Evaluation

**Software:** Autodesk Inventor • AutoCAD • Ansys (FEA) • MATLAB • Python MS • Office (Advanced Excel)

## PROFESSIONAL EXPERIENCE

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### Project Engineer | **Herrenknecht Tunnelling Services Panama Corp.**

*Panamá City, Panamá • 2025 – 2025*

- ▶ Supported engineering analysis and performance evaluation of mechanical systems across international TBM projects.
- ▶ Performed structural validation calculations and functional analysis of mechanical assemblies, contributing to safe operation and reduction of equipment downtime.
- ▶ Assisted in the development of technical documentation, SOPs, and engineering reports, improving execution consistency and reducing non-conformance events across 3+ project sites.
- ▶ Contributed to engineering decision-making by analyzing equipment behavior under operational conditions.

*Key projects: Hudson River Tunnel Project (USA) • Railway Tunnel Project (Perú)*

### Field Engineer – TBM & Surface Facilities (Intern) | **China Railway Tunnel Group Co., Ltd.**

*Panamá City, Panamá • 2024 – 2025*

- ▶ Supported mechanical system verification and performance analysis during tunnelling operations.
- ▶ Designed corrective engineering solutions for mechanical and piping systems, improving operational continuity and reducing field incidents.
- ▶ Produced P&IDs and isometric drawings for high-pressure systems in compliance with ASME B31 standards.
- ▶ Developed technical documentation and inspection procedures, reducing execution errors by 20%.
- ▶ Supported pressure testing, mechanical assembly verification, and system inspections.

*Key project: Panama Metro Line 3 – Tunnel Section*

### Engineering Projects – Mechanical Design & Analysis | **Independent**

*Panamá City, Panamá • 2024 – Present*

- ▶ Developed 3D mechanical designs and assemblies using Autodesk Inventor for structural and mechanical applications.
- ▶ Generated manufacturing drawings and technical documentation, applying dimensioning principles for fabrication.
- ▶ Designed pipe support systems, considering load conditions, material selection, and structural behavior.
- ▶ Applied engineering calculations to validate design assumptions and improve structural integrity.

## EDUCATION

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### **B.Sc. in Mechanical Engineering • Technological University of Panama**

*Panamá City, Panamá • Jan 2018 – Aug 2024*

**Relevant coursework:** Finite Element Analysis, Computational Modeling, Materials Engineering

## CERTIFICATIONS & TRAINING

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- Engineering training in hydraulic systems and industrial equipment monitoring – HK AG (Germany, Jun 2025)
- Design of Thermofluidic Systems – Technological University of Panama (Aug 2023)

## TECHNICAL PROJECTS & RESEARCH

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- Developed MATLAB-based model to analyze structural resonance under dynamic loading conditions.
- Applied numerical methods for engineering system evaluation.

## LANGUAGES

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**Spanish** (Native) • **English** (C1 – Full Professional Proficiency) • **German** (A2 – actively developing)

## FIELD & OPERATIONAL READINESS

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- Valid Driver License
- Work Authorization: Authorized to work in Panama
- Availability: Full-time, on-site